



PLMN Interactive Prototype

Public Land Mobile Network Explorer

Overview

PLMN Builder

Network Simulation

Components

What is PLMN?

A PLMN is a network established and operated by a mobile network operator to provide cellular services to mobile subscribers within a specific geographic area. It enables voice calls, SMS, and data connectivity for mobile devices.



Voice Calls

Make and receive calls



SMS

Send text messages



Data

Internet connectivity



HPLMN

Home Public Land Mobile Network

Your home network where your subscription is registered. Provides standard services at home rates.



VPLMN

Visited Public Land Mobile Network

A foreign network you connect to when roaming outside your home network area. May apply roaming charges.



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Build Your PLMN ID

MCC (Mobile Country Code)

440

3-digit country identifier

MNC (Mobile Network Code)

20

2-3 digit operator code

Your PLMN ID

440-20

Example PLMN Codes

US

AT&T
USA

310-410

GB

Vodafone
UK

234-15

DE

T-Mobile
Germany

262-01

JP

SoftBank
Japan

440-20



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Build Your PLMN ID

MCC (Mobile Country Code)

310

3-digit country identifier

MNC (Mobile Network Code)

410

2-3 digit operator code

Your PLMN ID

310-410

Example PLMN Codes

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MCC (Mobile Country Code)

234

3-digit country identifier

MNC (Mobile Network Code)

15

2-3 digit operator code

Your PLMN ID

234-15

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Build Your PLMN ID

MCC (Mobile Country Code)

262

3-digit country identifier

MNC (Mobile Network Code)

01

2-3 digit operator code

Your PLMN ID

262-01

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Network Connection Flow

▶ Start Simulation



Mobile device powers on and scans for networks



Device connects to nearest cell tower (BSS)



Device sends PLMN ID to identify the network



Network authenticates subscriber via HLR/VLR



Connection established and resources allocated



Services active: Voice, SMS, Data



✓ Connected to Network

All services are now available!

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Network Connection Flow

Pause Simulation

✓ Mobile device powers on and scans for networks

📶 Device connects to nearest cell tower (BSS)

📶 Device sends PLMN ID to identify the network

📶 Network authenticates subscriber via HLR/VLR

📶 Connection established and resources allocated

📶 Services active: Voice, SMS, Data

✓ Connected to Network
All services are now available!

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Network Connection Flow

⏏ Pause Simulation

- ✓

Mobile device powers on and scans for networks

>
- ✓

Device connects to nearest cell tower (BSS)

>
- ✓

Device sends PLMN ID to identify the network

>
- 

Network authenticates subscriber via HLR/VLR

>
- 

Connection established and resources allocated

>
- 

Services active: Voice, SMS, Data

>
- ✓

Connected to Network

All services are now available!

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Network Connection Flow

⏏ Pause Simulation

- ✓

Mobile device powers on and scans for networks

>
- ✓

Device connects to nearest cell tower (BSS)

>
- ✓

Device sends PLMN ID to identify the network

>
- ✓

Network authenticates subscriber via HLR/VLR

>
- ✓

Connection established and resources allocated

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Services active: Voice, SMS, Data

>

✓

Connected to Network

All services are now available!

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PLMN Architecture Components



Base Station Subsystem

Cell towers and base stations that communicate with mobile devices

Consists of Base Transceiver Station (BTS) and Base Station Controller (BSC).
Handles radio communication and manages radio resources.



Network Switching Subsystem

Manages call routing, mobility, and network services



Home Location Register

Central database storing subscriber information and profiles



Visitor Location Register

Temporary database for visiting subscribers in the network area

Component Interaction

All these components work together to provide seamless mobile connectivity:

BSS → NSS

NSS ↔ HLR

NSS ↔ VLR

VLR ↔ HLR

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PLMN Architecture Components



Base Station Subsystem

Cell towers and base stations that communicate with mobile devices



Network Switching Subsystem

Manages call routing, mobility, and network services

Core network component that includes MSC (Mobile Switching Center),
handles call setup, routing, and mobility management.



Home Location Register

Central database storing subscriber information and profiles



Visitor Location Register

Temporary database for visiting subscribers in the network area

Component Interaction

All these components work together to provide seamless mobile connectivity:

BSS → NSS

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